ARPANET (Advanced Research Projects Agency Network)

ARPANET (Advanced Research Projects Agency Network) was the world's **first packet-switching computer network** and the predecessor of the modern Internet. It was developed by the **Advanced Research Projects Agency (ARPA)** of the **U.S. Department of Defense (DoD)** in the late 1960s to enable reliable communication between geographically distributed computers.

Need for ARPANET

During the **Cold War** in the late 1950s and early 1960s, the U.S. Department of Defense wanted a communication network that could continue functioning even if parts of the network were destroyed during a military attack. The existing telephone network was centralized and vulnerable because the failure of a few switching centers could disrupt communication. To overcome this problem, ARPA initiated the development of a decentralized and fault-tolerant network called **ARPANET**.

Architecture of ARPANET

ARPANET consisted of three main components:

1. **Hosts** – Computers used by researchers and users.
2. **Interface Message Processors (IMPs)** – Specialized packet-switching computers (similar to modern routers).
3. **Communication Links – 56 kbps leased telephone lines** connecting the IMPs.

Working :

In **ARPANET**, when a host computer wants to send data, it first transmits the message to its local **Interface Message Processor (IMP)**. The IMP divides the message into smaller **packets** and forwards them through the network. Each packet travels independently and may follow different paths to reach the destination, depending on network conditions. At the receiving end, the destination IMP collects and reassembles all the packets in the correct order before delivering the complete message to the destination host. This **packet-switching** approach ensures reliable and efficient communication even if some network links fail. This technique is known as **store-and-forward packet switching**.

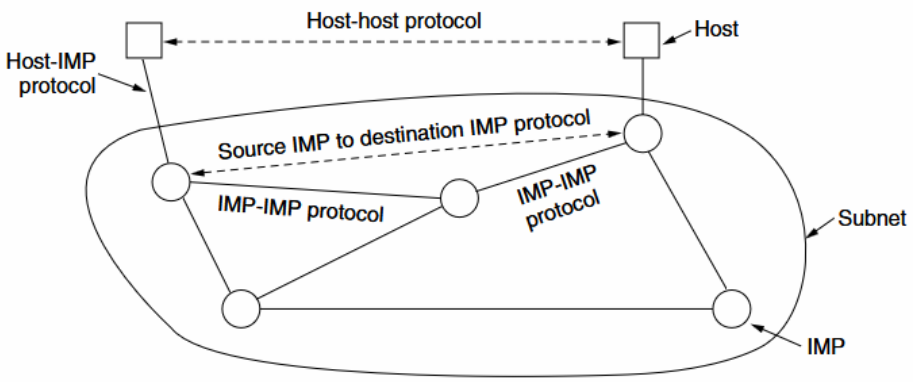


Figure 1-26. The original ARPANET design.

Features of ARPANET

- First **packet-switching network**.It used **distributed architecture**, eliminating a single point of failure.
- Supported **fault tolerance** by rerouting packets if a link failed. Used **Interface Message Processors (IMPs)** for packet forwarding.
- Connected heterogeneous computers from different manufacturers.
- Used **56 kbps communication links**.
- Enabled resource sharing among universities and research institutions.

Significance of ARPANET

ARPANET introduced the revolutionary concept of **packet switching**, which enabled data to be transmitted efficiently and reliably across computer networks. It demonstrated that **distributed networks** could continue functioning even if some communication links or network devices failed. The success of ARPANET led to the development and adoption of the **TCP/IP protocol suite**, which became the standard protocol for Internet communication. As the **foundation of the modern Internet**, ARPANET transformed the way computers communicate and revolutionized global communication, information sharing, and networking.

Internet

The **Internet** is a **global network of interconnected networks** that communicate using the **TCP/IP protocol suite**. It consists of Internet Service Providers (ISPs), routers, backbone networks, Internet Exchange Points (IXPs), and data centers, which work together to deliver data between users worldwide.

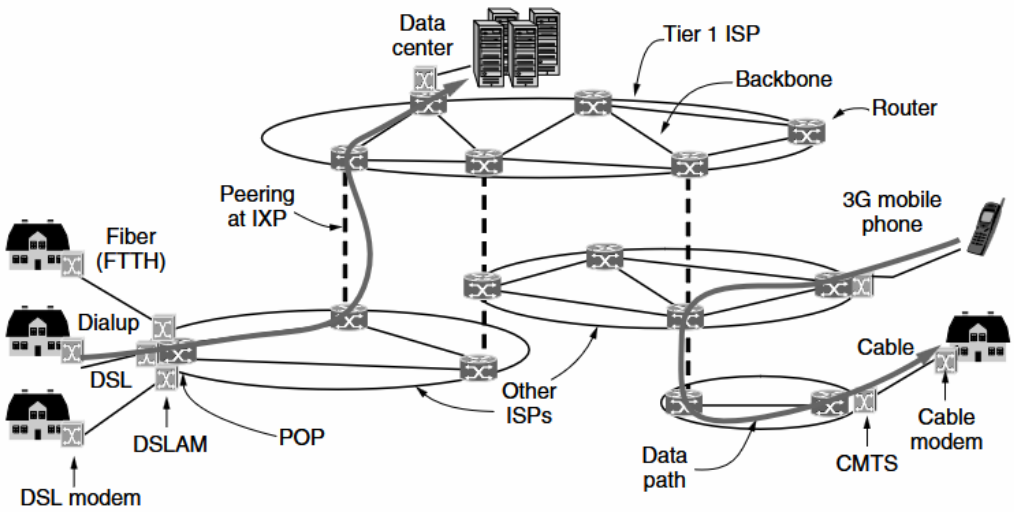


Figure 1-29. Overview of the Internet architecture.

Key Components of the Internet

- User Device
- Internet Service Provider (ISP)
- DSL / Cable / FTTH / Wireless Access
- Point of Presence (POP)
- ISP Backbone
- Internet Exchange Point (IXP)
- Tier-1 ISP
- Data Center / Server Farm

Architecture of the Internet

1. User Device : The Internet begins with a **user device** such as a computer, laptop, or smartphone. The device connects to the Internet through an **Internet Service Provider (ISP)**.

2. Internet Service Provider (ISP) : An **ISP** provides Internet connectivity to users. It allows users to send and receive data packets over the Internet.

3. Internet Access Technologies : Users can connect to an ISP using different technologies:

1. **DSL (Digital Subscriber Line) – Uses telephone lines.**
2. **Cable Modem – Uses cable TV networks.**
3. **Fiber to the Home (FTTH) – Uses optical fiber for high-speed Internet.**
4. **Wireless (Wi-Fi, 3G, 4G, 5G) – Provides Internet access without cables.**
5. **Dial-up Modem – Uses traditional telephone lines (older technology).**

4. Point of Presence (POP) : A **Point of Presence (POP)** is the location where customer data enters the ISP's network. It serves as the entry point to the Internet.

5. ISP Backbone : The **ISP backbone** is a high-speed network of routers and fiber-optic links that connects different cities and regions, carrying data across long distances.

6. Internet Exchange Point (IXP): An **Internet Exchange Point (IXP)** is a facility where different ISPs connect and exchange Internet traffic. This enables efficient communication between networks.

7. Tier-1 ISPs : **Tier-1 ISPs** are the largest global Internet providers. They own high-speed international backbone networks and connect all other ISPs, forming the core of the Internet.

8. Data Centers and Server Farms : Large companies such as **Google, Microsoft, and Amazon** store websites and online services in **data centers** containing thousands of servers. These server farms provide web pages, cloud services, email, and online applications.

Working of the Internet

When a user requests a web page, the request is sent from the user's device to the ISP through an access technology such as DSL, cable, fiber, or wireless. The ISP forwards the request through its backbone network and, if necessary, exchanges the traffic with other ISPs via Internet Exchange Points (IXPs). The request reaches the destination server located in a data center, which processes it and sends the requested information back to the user through the same Internet infrastructure.